

Dense span of convex nc functions in noncommutative Choquet theory

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Abstract

Davidson and Kennedy ask whether, for a compact noncommutative convex set K , the convex nc functions in $C(K)$ always span a dense subset of $C(K)$. We prove that the answer is yes. The proof is a Hahn–Banach consequence of Proposition 7.2.8 in the source paper: equality of unital completely positive maps on all matrix-valued convex nc functions forces equality of the maps.

1 Source question

The source is:

Kenneth R. Davidson and Matthew Kennedy, *Noncommutative Choquet theory*, arXiv:1905.08436.

The source asks the dense-span question immediately before Proposition 7.2.8:

... $C(K)$ is convex and only $C(K)$ is convex, so there is no disadvantage to working only with convex functions.

For a compact nc convex set K , we do not know if convex nc functions in $C(K)$ always span a dense subset of $C(K)$. However, the next result will be sufficient our purposes.

Proposition 7.2.8. *Let K be a compact nc convex set and let $\mu, \nu : C(K) \rightarrow \mathcal{M}$ be unital completely positive maps such that $\mu(f) = \nu(f)$ for every $n < \infty$ and every convex nc function $f \in \mathcal{M}_n(C(K))$. Then $\mu = \nu$.*

Proof. For $t \in [-1, 1]$, the function $h_t(x) = x^2(1 - tx)^{-1}$ is operator convex on the interval $(-1, 1)$ [5]. The Taylor series expansion of h_t at

We use the notation of the source paper. Thus K is a compact nc convex set, $C(K)$ is the C^* -algebra of continuous nc functions on K , and a convex nc function is self-adjoint.

2 Proof intuition

The source proposition says that convex nc functions separate u.c.p. maps, provided one tests all matrix-valued convex nc functions. To turn that into norm density, take a functional that annihilates the scalar convex functions. Its Jordan decomposition gives two states. Since vector compressions of matrix-valued convex functions are scalar convex functions, these two states also agree on every matrix-valued convex function. The source proposition then forces the states, and hence the original functional, to be equal to zero. Hahn–Banach finishes the density argument.

3 A compression lemma

Lemma 1. *Let $f \in M_n(C(K))$ be a self-adjoint convex nc function and let $\xi \in \mathbb{C}^n$. Define*

$$g_\xi(x) = (\xi^* \otimes 1)f(x)(\xi \otimes 1), \quad x \in K.$$

Then $g_\xi \in C(K)$ is a scalar-valued self-adjoint convex nc function.

Proof. Continuity, grading, direct sums, and unitary equivariance are inherited by compression. If $x = \sum_i \alpha_i^* x_i \alpha_i$ is an nc convex combination, convexity of f gives

$$f(x) \leq \sum_i (1_n \otimes \alpha_i^*) f(x_i) (1_n \otimes \alpha_i).$$

Compressing by $\xi \otimes 1$ gives

$$g_\xi(x) \leq \sum_i \alpha_i^* g_\xi(x_i) \alpha_i,$$

which is exactly convexity for the scalar nc function g_ξ . □

4 Dense span theorem

Theorem 1. *Let K be a compact nc convex set. The real linear span of the self-adjoint continuous convex nc functions in $C(K)$ is norm dense in $C(K)_{\text{sa}}$. Consequently, the complex linear span of the continuous convex nc functions is norm dense in $C(K)$.*

Proof. Let E be the real linear span of the scalar-valued self-adjoint continuous convex nc functions in $C(K)$. Suppose that $\overline{E} \neq C(K)_{\text{sa}}$. By the real Hahn–Banach theorem, there is a non-zero bounded real linear functional ϕ on $C(K)_{\text{sa}}$ such that $\phi|_E = 0$.

Since the constant function 1 is affine, hence convex, we have $\phi(1) = 0$. Let

$$\phi = \phi_+ - \phi_-$$

be the Jordan decomposition of the bounded self-adjoint functional ϕ on the C^* -algebra $C(K)$. The equality $\phi(1) = 0$ gives $\phi_+(1) = \phi_-(1)$. If this common value is zero, then $\phi = 0$, a contradiction. Otherwise set

$$\mu = \phi_+ / \phi_+(1), \quad \nu = \phi_- / \phi_-(1).$$

These are states on $C(K)$. Because ϕ vanishes on every scalar convex nc function, μ and ν agree on every scalar convex nc function.

We now show that μ and ν agree on every matrix-valued convex nc function. Let $f \in M_n(C(K))$ be self-adjoint and convex. For each $\xi \in \mathbb{C}^n$, Lemma 1 gives a scalar convex nc function g_ξ . Hence

$$\xi^* (\mu_n(f) - \nu_n(f)) \xi = \mu(g_\xi) - \nu(g_\xi) = 0.$$

Since $\mu_n(f) - \nu_n(f)$ is self-adjoint and all of its quadratic forms vanish, $\mu_n(f) = \nu_n(f)$.

Thus μ and ν satisfy the hypothesis of Proposition 7.2.8 of Davidson–Kennedy: they agree on every $n < \infty$ and every convex nc function in $M_n(C(K))$. Therefore $\mu = \nu$. It follows that $\phi = 0$, contradicting the choice of ϕ . Hence $\overline{E} = C(K)_{\text{sa}}$.

Every element of $C(K)$ is $a + ib$ with $a, b \in C(K)_{\text{sa}}$, so the complex linear span of the continuous convex nc functions is dense in $C(K)$. □

5 Verification and novelty notes

The argument uses only standard Hahn–Banach separation, the Jordan decomposition of bounded self-adjoint functionals on a C^* -algebra, and Proposition 7.2.8 of the source paper. No computational verification is needed.

Local run indexes had no prior packet for arXiv:1905.08436. A bounded search on 2026-07-18 for the exact dense-span formulation found the source paper and the later survey arXiv:2412.09455, but no explicit statement of Theorem 1. The result should be understood as a short corollary of the source paper’s own separation proposition.

6 Limitations

This proves density of the linear span. It does not prove the stronger exact decomposition statement that every element of $C(K)_{\text{sa}}$ is literally the difference of two continuous convex nc functions.

7 Human review recommendation

Review as a likely valid full solution. The central point to verify is the compression step: scalar vector compressions of matrix-valued convex nc functions remain scalar convex nc functions, allowing Proposition 7.2.8 to be applied to the normalized positive and negative parts of a separating functional.

References

- [1] K. R. Davidson and M. Kennedy, *Noncommutative Choquet theory*, arXiv:1905.08436; Mem. Amer. Math. Soc. 316 (2025), no. 1608.
- [2] K. R. Davidson and M. Kennedy, *Noncommutative Choquet theory: A survey*, arXiv:2412.09455, 2024.